undisturbed base of the organic sediments in these cases shows that the pingo ice had already melted out and that the top of the grey clay-silt was stable before the end of Pollen Zone III.

The date of their formation remains open; but the significance of the pingos for the age of the latest deglaciation of Cardiganshire, perhaps, lies more in the necessity for a time gap between deglaciation and the possibility of pingo formation, as is shown by a recent study of the relationship of contemporary pingos to glacial limits in the Yukon[12]. In the central and southern Yukon, two glacial limits are indicated by moraines outside the Neoglacial moraines associated with the snouts of modern glaciers. These belong to the McConnell Glaciation, dated by overlying organic silt to >12,000 BP and equated with the Wisconsin, and the Reid Glaciation dated by overlying organic material as >42,900 BP and therefore Early Wisconsin or Pre-Wisconsin in age[13]. Outside the Reid limit there are glacial deposits "lacking distinctive geomorphic form" (ref. 13), and beyond these the unglaciated part of the Yukon Plateau. Hughes[12] found that pingos are most abundant in the unglaciated region and "extend into the areas of Pre-Reid Glaciation without any sharp decrease in density". Density is, however, "markedly lower in areas that were last glaciated during the Reid advance", while "only two definite and two probable pingos have been identified within the limits of the McConnell advance".

Yet present day pingos in Canada and Alaska are believed to have been forming continuously for several thousand years, probably for the past 4,000 yr and just possibly for the past 7,000 yr (ref. 8). Their virtual absence in the considerable area which was covered by the McConnell advance but has been ice-free for more than 7,000 yr makes it clear that the growth of pingos has not followed on the retreat of the ice even in conditions of discontinuous permafrost. Hughes[12] also suggests that the low density of pingos in the areas of the Reid Glaciation may be attributed to the relative freshness of the glacial drift morphology and materials. For the open system pingos of the Yukon basin are mostly sited on slope deposits, "on slopes mantled by several hundred feet of slopewash sediments and colluvial rock debris" (ref. 12). In central Alaska, "none have been found on glacial drift" (ref. 8).

The evidence of the present day open system pingos of central Yukon suggests that rare examples may occur inside the limit of the last glaciation, but that high densities are found only outside it. The pingos of Cardiganshire are not sparse in distribution but occur in a number of localities where they are found in considerable profusion suggesting that several generations have developed. These pingo groups are sited in open valleys or basins where the long regular slopes indicate that slope processes have been long dominant. The frequent worn and rounded pebbles of local rock in the slope deposits east of the Ceri basin (Fig. 1), and the northern erratic pebbles and Irish Sea Till material in the Ceri and Hirwaun basins, show that these slope processes have transformed a landscape of glacial deposition. The pingos involve these slope deposits.

Charlesworth believed that the sand and gravel deposits around the Teifi estuary and those of the upper Teifi below Tregaron marked the ice limit of the last glaciation and drew a line between them, passing around the northern end of the Mynydd Bach (Fig. 1). To the south of the Aeron, this was based on limited evidence for ice margin drainage, but north of this the evidence was "singularly scanty" and he described his line as "conjectural" (ref. 2). Wirtz interpreted the evidence around the Teifi estuary and further south-west as indicating the limit of the Irish Sea Ice of the last glaciation, the area to the east and north-east being ice free[2]. This seems more in accord with the distribution of pingos which are well developed both inside and outside Charlesworth's "Newer Drift" limit.

EDWARD WATSON

Department of Geography,
University College of Wales,
Aberystwyth, Cardiganshire

Received January 13, 1972.

1 Charlesworth, J. K., *Quart. J. Geol. Soc. Lond.*, **85**, 335 (1929).
2 Wirtz, D., *Neues Jb. Geol. Paläont.*, **96**, 267 (1953).
3 Mitchell, G. F., *Adv. Sci.*, **17**, 313 (1960).
4 Synge, F. M., *Welsh Soils Discussion Group*, **2**, 15 (1961).
5 Johns, B. S., and Ellis-Gruffydd, I. D., *Geol. en Mijnbouw*, **49**, 285 (1970).
6 Watson, E., *Geol. J.*, **7**, 381 (1971).
7 Pissart, A., *Z. Geomorphol.*, **7**, 147 (1963).
8 Holmes, G. W., Hopkins, D. M., and Foster, H. L., *US Geol. Surv. Bull.*, 1241-H (1968).
9 Watson, E., and Watson, S., *Rep. Twenty-fourth Intern. Geol. Congr. Canada* (in the press).
10 Trotmann, D. M., *Z. Geomorphol.*, **7**, 168 (1963).
11 Slotboom, R.-T., *Z. Geomorphol.*, **7**, 165 (1963).
12 Hughes, O. L., *Geol. Surv. Canada*, Paper 69–34 (1969).
13 Hughes, O. L., Campbell, R. B., Muller, J. E., and Wheeler, J. D., *Geol. Surv. Canada*, Paper 68–34 (1969).

Palaeomagnetism and the Origin of the Grenville Front

THE Grenville Province is bounded on the north by the Grenville Front (Fig. 1) and on the south-east by the Appalachian Fold Belt. To the south-west in the St Lawrence Lowlands, Grenville rocks pass beneath sub-horizontal Lower Palaeozoic cover rocks. The Grenville Front is observed to be a fault or metamorphic boundary extending for 1,500 km which truncates several older structural trends. It separates Archaean and Lower and Middle Proterozoic rocks (some of which are unmetamorphosed) to the north from more strongly metamorphosed but generally younger rocks to the south.

According to the theory of plate tectonics, there are three types of plate junctures named by the motions that occur across them—divergent, convergent, and strike-slip. The Grenville Front has no large linear belts of basic rocks such as characterize divergent or convergent junctures. Moreover, the metamorphic trends (they are predominantly northerly[1]) in the Province generally are not parallel to the Front. Thus plate theory would suggest that the Grenville Front is a strike-slip juncture formed after the main metamorphism of Grenville rocks. In this article, we shall summarize the palaeomagnetic evidence for this idea.

The Grenville Province[2] consists of (1) basement gneiss complexes; (2) large anorthosite-mangerite plutons; and (3) smaller intrusions of gabbro, diorite, syenite and granite, all

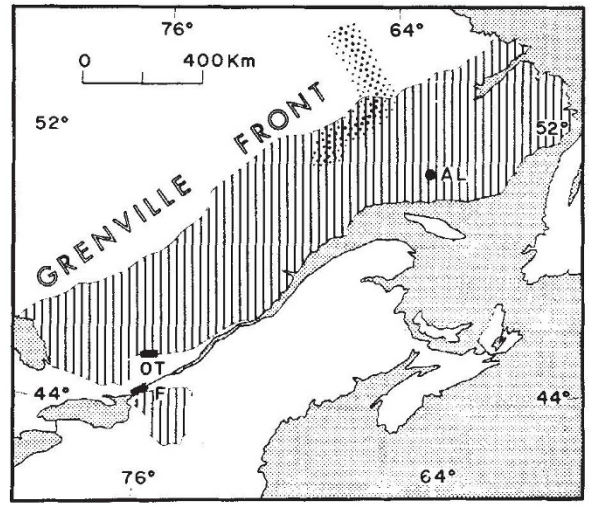


Fig. 1 The Grenville Province is marked by vertical shading. AL is the Allard Lake Anorthosite, OT the Ottawa area basic intrusives, and F the north-west dykes of the Frontenac axis. The dotted area shows the continuation of the Labrador Trough rocks (Lower Proterozoic) into the border zone of the Grenville Province and their clockwise deflexion at the Front.

© 1972 Nature Publishing Group

of which have undergone moderate to high-grade regional metamorphism. Following this intense metamorphism small granite and syenite bodies (4) and basic dykes (5) were emplaced. Elements (1) to (4) yield K-Ar dates with an average of 950 m.y., but older dates, varying from 1,700 to 1,100 m.y., are obtained by the Rb-Sr and U-Pb methods. The basic dykes have a variety of trends and ages. One prominent set (the so-called Grenville dykes[3]) are late Precambrian (about 675 m.y.); they are probably related to the early rifting stages of the Palaeozoic diastrophic cycle that created the Appalachian Fold Belt and are therefore not considered here.

Early palaeomagnetic studies of element (1) (Russell Border Belt[4] of the Adirondacks, declination 145, inclination +50), of gabbros[5] of element (3) (Thanet 093, +62 precision[6] $k=5$, Tudor 328, +11 $k=5$, and Umfraville 115, +43 $k=8$), and of various rocks (mean direction 287, −44) related to element (3), yielded magnetization directions aligned along east-south-east to west-north-west axis. Both normal and reversed polarities are present, and the scatter of directions is high, but detached demagnetization studies of rocks from three places have recently produced results with much less internal scatter, the first[7] (from element 2), from the Allard Lake anorthosite of eastern Quebec (AL of Fig. 1, mean direction 297, −79 $k=88$ pole 39S, 140E $A_{95}=18$). The second result (Park and Irving, unpublished) is from six north-west trending basic dykes of the Frontenac axis near Kingston (F of Fig. 1, mean direction 103, +50 $k=112$ pole 12S, 163E $A_{95}=7°$). K-Ar whole-rock[18] and biotite[9] ages of 751 and 817 m.y. have been reported from these dykes and presumably indicate their minimum age.

A third result from basic intrusions of element (3) in the Ottawa area is now briefly described. Two bodies situated about 50 km west of Ottawa were sampled. One intrusion is near Bryson (45° 41′ N, 076° 37′ W; five sites) and the other is at Cheneaux Falls on the Ottawa River (45° 35′ N, 076° 40′ W; one site). In another area, 20 km north of Ottawa in the Gatineau Valley, several small bodies marginal to the Wakefield syenite were sampled (45° 37′ N, 075° 51′ W; eight sites). All are dioritic or gabbroic in composition and are foliated to varying degrees. At each site five to seven core samples were drilled (seventy in all). Two cylindrical specimens were cut from each core. Detailed alternating field (a.f.) demagnetization studies were made of one specimen from each core and an end-point determined. Thirteen specimens had no stable end-points. For the other fifty-seven specimens stable end-points were found in the field range of 150 to 300 Oe. The second specimen from each core was then demagnetized in this field. Eight sites have reversed and five have normal magnetization (Table 1). The corresponding poles for this and the other Grenville results are situated near Australia (Fig. 2) and fall in a sequence corresponding to their supposed ages—anorthosite (AL element 2), minor plutons (OT element 3), and north-west trending dykes (F element 5).

Table 1 Summary of Results

	D, I	N	k	α_{95}
B04	301, −63	4	33	16
B04 (M)	296, −61	6	22	12
B05	269, −59	4	94	10
B11	278, −69	5	47	11
B (Ave)	286, −64	4	99	09
C13	109, +64	4	14	26
G01	266, −70	7	761	02
G20	104, +59	4	11	29
G19	068, +69	9	9	18
G18 (N)	041, +62	4	57	12
G18 (R)	267, −74	5	633	03
G17	266, −61	5	57	10
G27	070, +67	3	71	15
G26	276, −61	3	17	31
G (Ave)	260, −67	8	65	07
Total	271, −66	13	61	05
Pole	32S, 155E	13	25	08

The B sites are from the Bryson body, B04M being data of Murthy[3]. The C site is from Cheneaux. The G sites are from the Gatineau Valley. The site numbers are laboratory catalogue numbers and have no geological significance. D, I are declination and inclination, N the number of cores (2 specimens from each) or number of sites in the averages, k is Fisher's precision, and α_{95} the radius of the circle of confidence ($P=0.05$). At one location (G18) two irregular dyke-like bodies have opposite polarity.

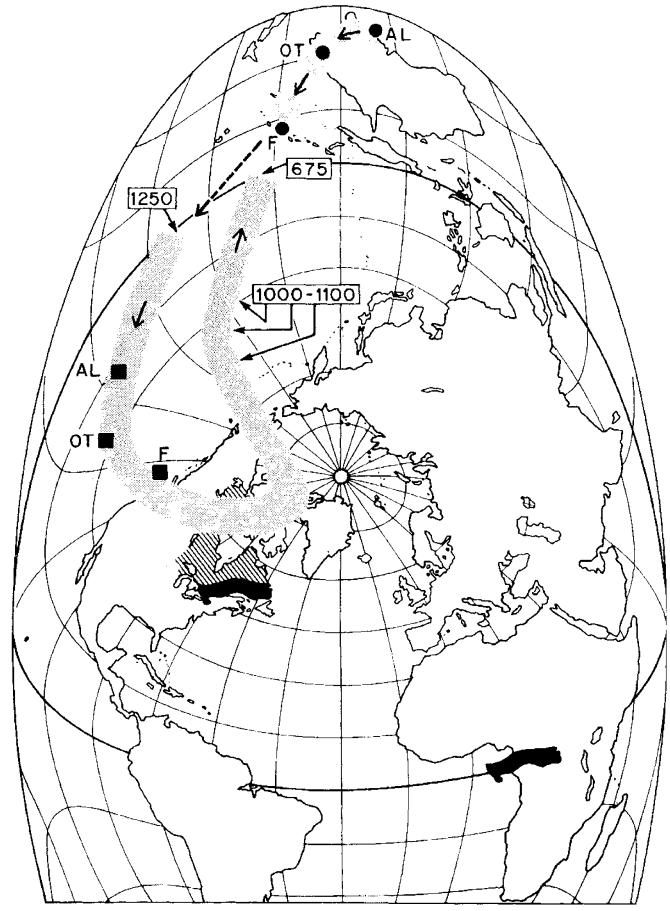


Fig. 2 The polar sequence AL–OT–F is obtained from Grenville rocks (see legend to Fig. 1 and text). The Grenville Province is black. The large loop is the Proterozoic polar path relative to the rest of the Canadian Shield which is shaded. The sense of polar motion in the loop is denoted by arrows and the terminal ages are given in millions of years. This loop was first roughed out by Du Bois[17] and has since been further developed by Spall[18] and by Robertson and Fahrig[19]. The loop drawn here is deeper than that of earlier authors and has been constructed using a recent compilation of Precambrian results (unpublished work). The polar motions depicted are relative to the Canadian Shield only; the continental outlines are added to provide a frame of reference. The proposed position of the Grenville Province (in what is nowadays West Africa) before the formation of the Grenville Front has been obtained graphically by moving the Grenville poles (now plotted as squares) into the descending limb of the loop and moving the Grenville Province accordingly. The uncertainties in this procedure are discussed in the text.

The path of apparent polar wander relative to the rest of the Canadian Shield for the interval between the Mackenzie[10–12] (1,250 m.y.) and Franklin[13] (675 m.y.) igneous episodes is also plotted in Fig. 2. This path has the form of a loop on which the Grenville poles do not fall, which could mean that the late Proterozoic polar path is much more complicated than is depicted in Fig. 2. For example, the Grenville poles could represent an early section of the descending limb of the polar loop as indicated by the dashed line. (This is an improbable reconstruction since the dykes of pole F are almost certainly younger than the Mackenzie diabases, the starting point for the polar loop, and the poles from rocks in the age range 1,400–1,300 m.y. fall in the east-central Pacific, not near Australia.) Alternatively, we may suppose that the Grenville Province was

© 1972 Nature Publishing Group

once displaced relative to the remainder of the Canadian shield[14,15].

This hypothesis has several interesting consequences. The first is that the magnetizations of the Allard Lake anorthosite and the Ottawa basic rocks were acquired about 1,200 m.y. ago presumably during cooling after the last penetrative metamorphism; that is the magnetizations "see through" the K-Ar event at 950 m.y. The second consequence is that at the time of metamorphism, the Grenville Province was situated many thousands of kilometres distant from the older Canadian Shield, its position relative to the Shield (Fig. 2) being such that motion required to bring them together is predominantly dextral strike-slip. A component of convergence perpendicular to the front may have been present and its magnitude determined by this method depends on where the Grenville poles are placed on the descending limb of the polar loop. Although the data allow considerable freedom, whatever placing is adopted dextral strike-slip predominates, and relative motion is of the order of several thousands of kilometres.

This idea, that the Grenville Front is a strike-slip juncture along which motions comparable in scale with current plate motions occurred, could explain many observed features. It could explain the zones of cataclasis and mylonitization along the front that are believed to have developed during the later stages of deformation, after the main metamorphic episode[2]. It could explain how generally older, sometimes unmetamorphosed rocks to the north have been brought into such close contact with the younger more highly metamorphosed terrain of the Grenville Province. It could explain the presence of a border zone of mixed rocks south of the Front, some of which (notably the extension of the basic rocks and iron formation of the Labrador Trough) swing westward as if they have been rotated clockwise (Fig. 1).

We suggest that the time of collision of the two continental masses is marked by the peak of the polar loop; more specifically that the rapid apparent polar change that occurs in certain Keweenawan rocks whose age is indistinguishable radiometrically[16] actually corresponds to and is caused by this collision. On this view the collision occurred between 1,200 to 1,100 m.y. ago, and the K-Ar dates (average 950 m.y.) then correspond to a period of vertical adjustment (general uplift) consequent on this collision.

Finally, we note that the predominantly strike-slip juncture of the present day Pacific and North American plates has relatively small divergent sections at points where strain is transferred from one strike-slip fault system to another. By analogy we suggest that the Keweenawan Basin and the Naskaupi Fold Belt in Labrador were developed as tensional rifts (basic intrusions and lavas and thick clastic sediments) associated with geometrical irregularities in the developing Grenville Front.

The graphical operations depicted in Fig. 2 were carried out on a large globe (scale 1 cm = 1°) using spherical plastic overlays, accuracies of ¼° being readily attainable. This globe was constructed by John Geuer and Odette Degen.

E. IRVING
J. K. PARK
J. L. ROY

Earth Physics Branch,
Department of Energy, Mines and Resources, Ottawa

Received January 26, 1972.

1 Stockwell, C. H., in *Geol. Econ. Mins. Canada* (edit. by Douglas, R. J. W.), 44 (Dept. Energy, Mines and Resources, Ottawa, 1970).
2 Emslie, R., *Geol. Econ. Mins. Canada* (edit. by Douglas, R. J. W.), 121 (Dept. Energy, Mines and Resources, Ottawa, 1970).
3 Murthy, G. S., *Canad. J. Earth Sci.*, **8**, 802 (1971).
4 Balsley, J. R., and Buddington, A. F., *Phil. Mag. Supp. Adv. Phys.*, **6**, 317 (1957).
5 Hood, P. J., *J. Geophys. Res.*, **66**, 1235 (1961).
6 Fisher, R. A., *Proc. Roy. Soc.*, A, **217**, 295 (1953).
7 Hargraves, R. B., and Burt, D. M., *Canad. J. Earth Sci.*, **4**, 357 (1967).
8 Wanless, R. K., Stevens, R. D., Lachance, G. R., and Rimsaite, J. G. H., *Geol. Survey Canada Paper*, 65–17, 75 (1965).
9 Wanless, R. K., Stevens, R. D., Lachance, G. R., and Edmonds, C. M., *Geol. Survey Canada Paper*, 66–17, 88 (1966).
10 Robertson, W. A., *Bull. Geol. Surv. Canada*, **167**, 52 (1969)
11 Fahrig, W. F., and Jones, D. L., *Canad. J. Earth Sci.*, **6**, 679 (1969).
12 Irving, E., Park, J. K., and McGlynn, J. C., *Canad. J. Earth Sci.* (in the press).
13 Fahrig, W. F., Irving, E., and Jackson, J. D., *Canad. J. Earth Sci.*, **8**, 455 (1971).
14 Irving, E., *Geofis. Pura. Appl.*, **33**, 23 (1956).
15 Runcorn, S. K., *Proc. Geol. Assoc. Canada*, **8**, 77 (1956).
16 Palmer, H. C., *Canad. J. Earth Sci.*, **7**, 1410 (1970).
17 Du Bois, P. M., *Geol. Surv. Canada Bull.*, **71**, 1 (1962).
18 Spall, H., *Earth Planet. Sci. Lett.*, **10**, 273 (1971).
19 Robertson, W. A., and Fahrig, W. F., *Canad. J. Earth Sci.*, **8**, 1355 (1971).

BIOLOGICAL SCIENCES

In situ Hybridization of Adenovirus RNA and DNA

THE methods which have been developed for annealing nucleic acids *in situ*[1,2] have recently been utilized in locating the positions of satellite fractions of DNA in the cells of a number of species including mouse[3,4] and man[5]. RNA complementary to viral DNA can be used in a similar manner to localize viral nucleic acid in the host cell[6].

Adenoviruses have been shown to induce chromosome aberrations in both permissive and non-permissive cells[7–9], and in the case of the three serotypes highly oncogenic in hamsters[10] to induce non-random chromosome damage to human cells *in vitro*[11,12]. In this report we describe our initial results using adenovirus type 12 complementary RNA (cRNA) to follow virus replication in human cells and to examine the relationship between the viral genome and the host cell chromosomes.

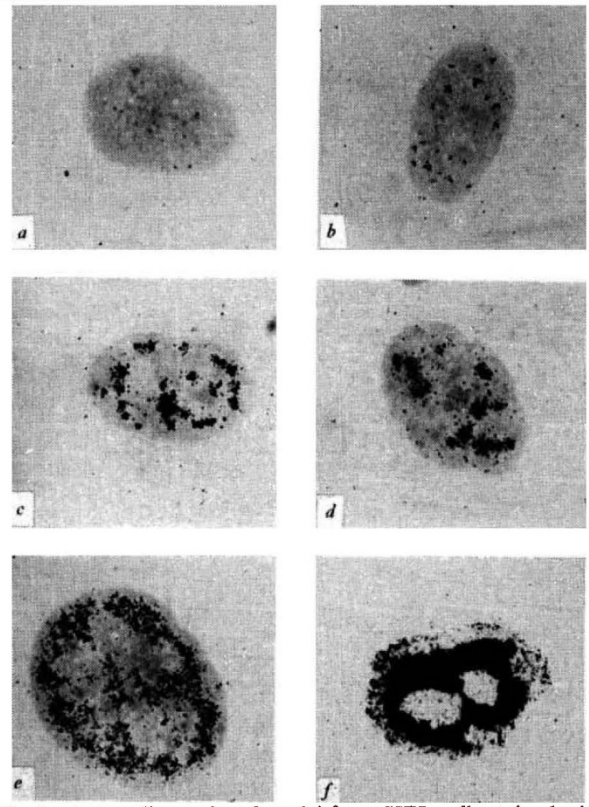


Fig. 1 Autoradiographs of nuclei from HEL cells stained with 'Giemsa' after 8 weeks exposure. *a*, Uninfected cell; *b–f*, progressive stages of viral DNA synthesis detected by *in situ* hybridization of adenovirus cRNA (×480).

© 1972 Nature Publishing Group